DATA SCIENCE & ARTIFICIAL INTELLIGENCE

& CS/IT

Calculus and Optimization

Taylor and Maclaurin series

One Shot



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Recap of previous lecture



Topic BASICS of LIMITS & CONTINUITY

Topics to be Covered



Topic

TAYLOR & MACLAURIN SERIES

(POWER SERIES)

Q. $\lim_{x\to 0}\left(\frac{a^x+b^x+c^x}{3}\right)^{\frac{1}{x}} = ? \approx \left(\frac{1+1+1}{3}\right)^{\frac{1}{0}} = (1)^{\infty}$ form.

(a) $abc/3$

(b) ✓ $(abc)^{1/3}$

(c) $3abc$

(d) 1

$$K = \lim_{x\to 0}\left(\frac{a^x+b^x+c^x}{3}\right)^{\frac{1}{x}}$$

$$\log_e K = \lim_{x\to 0} \log\left\{\frac{a^x+b^x+c^x}{3}\right\}^{1/x} = \lim_{x\to 0}\frac{\log\left(\frac{a^x+b^x+c^x}{3}\right)}{x} = \frac{0}{0}$$

$$\log_e K = \lim_{x\to 0}\frac{\frac{1}{\left(\frac{a^x+b^x+c^x}{3}\right)}\left(\frac{a^x\ln a+b^x\ln b+c^x\ln c}{3}\right)}{1}$$

$\because \frac{d}{dx}\log X = \frac{1}{X}$

$\frac{d}{dx}(a^x) = a^x \log_e a$

$$\log_e K = \lim_{x \to 0} \left(\frac{a^x \ln a + b^x \ln b + c^x \ln c}{a^x + b^x + c^x} \right) = \frac{\ln a + \ln b + \ln c}{1+1+1} = \frac{\ln(abc)}{3}$$

$$\ln K = \frac{1}{3} \ln(abc) = \ln(abc)^{1/3}$$

$$\Rightarrow K = (abc)^{1/3}$$

(HW): $\lim_{x\to 0} \dfrac{(4^x-1)^3}{\sin\left(\frac{x}{4}\right)\cdot \log\left(1+\frac{x^2}{3}\right)} = ?$

(M-I): Irritating

(M-II) Using standard Results.

$$= \lim_{x\to 0} \frac{\left(\frac{4^x-1}{x}\right)^3 \times x^3}{\frac{\sin\left(\frac{x}{4}\right)}{\left(\frac{x}{4}\right)} \cdot \frac{\log\left(1+\frac{x^2}{3}\right)}{\left(\frac{x^2}{3}\right)} \times \left(\frac{x}{4}\times\frac{x^2}{3}\right)}$$

$\because \lim_{x\to 0}\left(\frac{a^x-1}{x}\right) = \ln a$

$\lim_{x\to 0}\frac{\log(1+x)}{x} = 1$

$$= \frac{\left[\lim_{x\to 0}\left(\frac{4^x-1}{x}\right)\right]^3 \times 12}{\lim_{\frac{x}{4}\to 0}\frac{\sin\left(\frac{x}{4}\right)}{\left(\frac{x}{4}\right)} \cdot \lim_{\frac{x^2}{3}\to 0}\frac{\log\left(1+\frac{x^2}{3}\right)}{\left(\frac{x^2}{3}\right)}} = \frac{(\ln 4)^3 \times 12}{1\times 1} = 12(\ln 4)^3$$

Q: If $2-\frac{x^2}{3} < \frac{x\sin x}{1-\cos x} < 2$ then Evaluate $\lim_{x\to 0}\left(\frac{x\sin x}{1-\cos x}\right) = ?$

M-I $\lim_{x\to 0}\left(\frac{x\sin x}{1-\cos x}\right) \simeq \frac{0}{0} = \lim_{x\to 0}\left(\frac{x\cos x + \sin x}{+\sin x}\right) = \frac{0}{0}$ form.

$$= \lim_{x\to 0} \frac{x(-\sin x)+\cos x+\cos x}{\cos x} = \frac{0+1+1}{1} = 2 \quad \text{Ans}$$

M-II Using SANDWITCH Theorem →

if $f(x) < g(x) < h(x)$ then

$\lim_{x\to a} f(x) < \lim_{x\to a} g(x) < \lim_{x\to a} h(x)$

$\because \lim_{x\to 0}\left(2-\frac{x^2}{3}\right) = 2$

& By S.T., $\lim_{x\to 0}\left(\frac{x\sin x}{1-\cos x}\right) = 2$

Q. $\lim_{x\to\infty}\left(\frac{3x-8\sin x}{2x+5\cos x}\right)=?=\lim_{x\to\infty}\frac{3-\left(\frac{8\sin x}{x}\right)}{2+5\left(\frac{\cos x}{x}\right)}$

(a) 3

(b) 1.5

(c) 1

(d) 0

$$=\frac{3-0}{2+5(0)}=\frac{3}{2}$$

TAYLOR & MACLAURIN SERIES

(MACLAURIN SERIES) → Let $f(x)$ is Continuous func^n defined in a Domain D s.t it's all derivative exist then $f(x)$ can be expanded in the Nbd of 0 $(x \to 0)$ as follows;

$$f(x) = f(0) + x f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \cdots\cdots\cdots$$

Note ① this series provides Approx value of $f(x)$ in the Nbd of 0

② Coeff of x^n in the M.S.Exp $= \dfrac{f^n(0)}{n!}$

③ In this series we are getting Increasing powers of x

Some Standard Results of Maclaurin Series → All these Results are valid only when $x \to 0$

(1) $\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \frac{x^5}{5} - \cdots\cdots$

$\log(1-x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} - \frac{x^5}{5} - \cdots\cdots$

(2) $a^x = 1 + x(\ln a) + \frac{x^2}{2!}(\ln a)^2 + \frac{x^3}{3!}(\ln a)^3 + \cdots\cdots$

Gate ✓ $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots\cdots$

(3) $\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \frac{x^9}{9!} - \cdots\cdots$

$\sinh x = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \frac{x^7}{7!} + \cdots\cdots = \frac{e^x - e^{-x}}{2}$

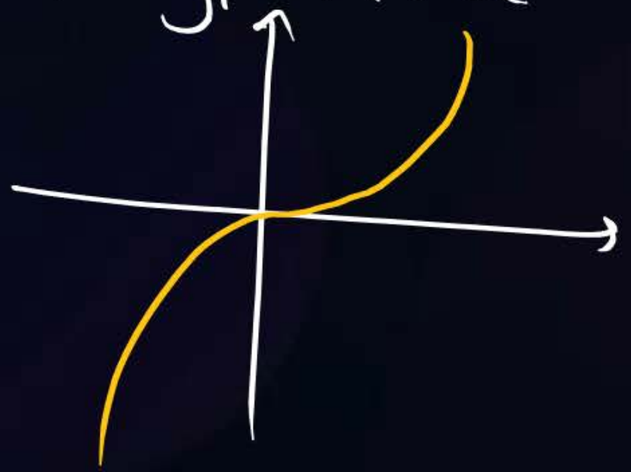
(4) $\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \frac{x^8}{8!} - \cdots\cdots\cdots\infty$

$\cosh x = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \frac{x^6}{6!} + \cdots\cdots\cdots\cdots = \frac{e^x + e^{-x}}{2}$

(5) $\tan x = x + \frac{x^3}{3} + \frac{2}{15}x^5 + \cdots\cdots\cdots$ (Proove it / HW)

Note: $\sinh x = \frac{e^x - e^{-x}}{2}$,

$\frac{d}{dx}(\sinh x) = \int \sinh x\,dx = \cosh x$



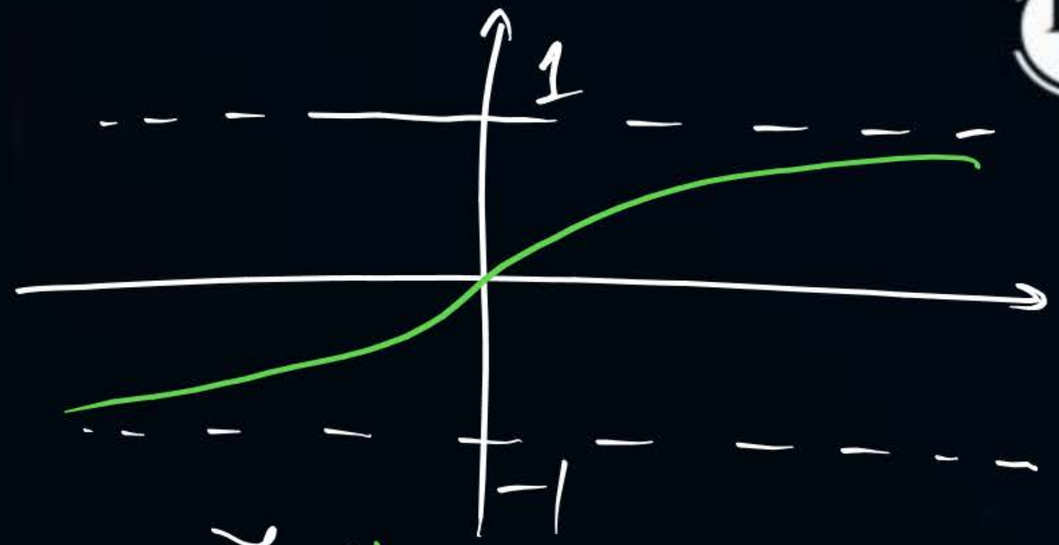
$\cosh x = \frac{e^x + e^{-x}}{2}$

$\frac{d}{dx}(\cosh x) = \int \cosh x\,dx = \sinh x$

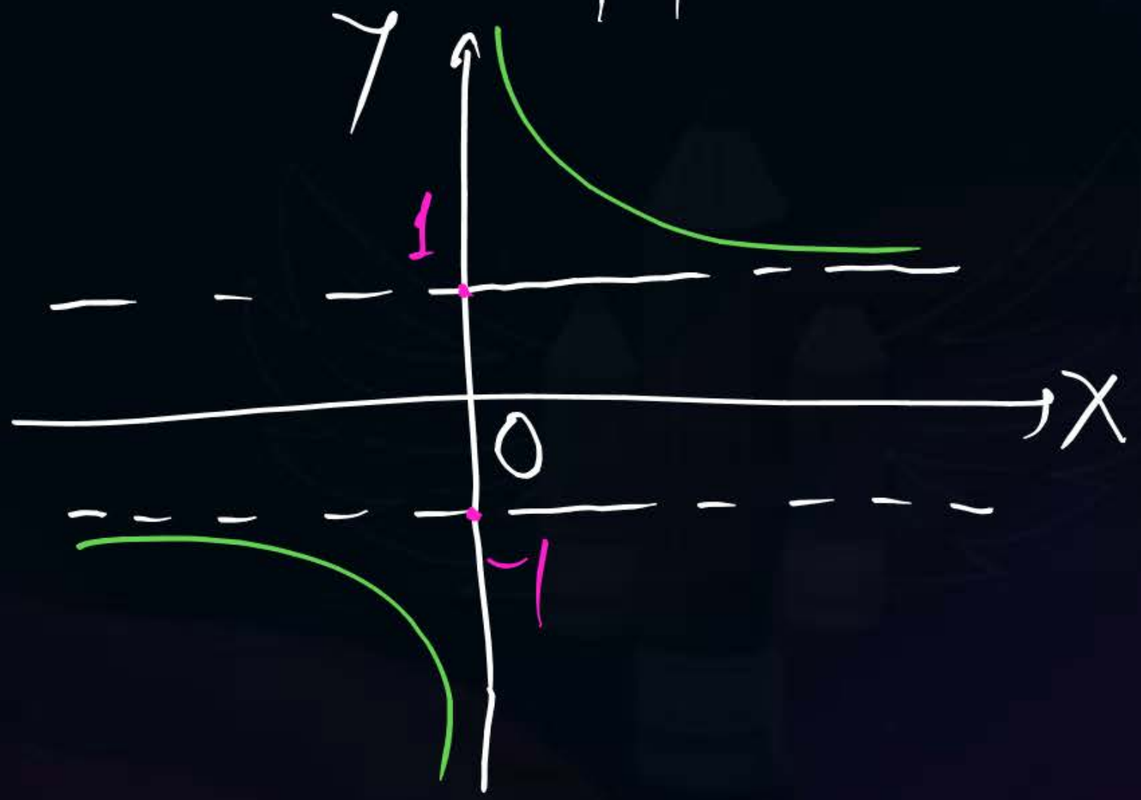


Note:

① $\tanh x = \dfrac{\sinh x}{\cosh x} = \dfrac{e^x - e^{-x}}{e^x + e^{-x}}$



② $\coth x = \dfrac{\cosh x}{\sinh x} = \dfrac{e^x + e^{-x}}{e^x - e^{-x}}$



Q: Which of the following series consist only odd powers of x in its Taylor series expansion about $x=0$?

GATE

(a) $\sin(x^2) = (x^2) - \frac{(x^2)^3}{3!} + \frac{(x^2)^5}{5!} - \cdots\cdots\cdots$

(b) $\cos(x^2) = 1 - \frac{(x^2)^2}{2!} + \frac{(x^2)^4}{4!} - \frac{(x^2)^6}{6!} + \cdots\cdots$

(c) $\sin(x^3) = (x^3) - \frac{(x^3)^3}{3!} + \frac{(x^3)^5}{5!} - \frac{(x^3)^7}{7!} + \cdots\cdots$

(d) $\cos(x^3) = 1 - \frac{(x^3)^2}{2!} + \frac{(x^3)^4}{4!} + \frac{(x^3)^6}{6!} + \cdots\cdots$

Q: Expand $\log(1+x)$ in the Increasing powers of x ?

M.S. Exp.

Sol: $f(x)=\log(1+x)$, $f'(x)=\frac{1}{1+x}$, $f''(x)=\frac{-1}{(1+x)^2}$, $f'''(x)=\frac{2}{(1+x)^3}$ .....

$f(0)=\log(1+0)$, $f'(0)=\frac{1}{1+0}$, $f''(0)=\frac{-1}{(1+0)^2}$, $f'''(0)=\frac{2}{(1+0)^3}$

$=0$ $=1$ $=-1$ $=2$

Now Standard Result of Maclaurin Series is:

$$f(x)=f(0)+xf'(0)+\frac{x^2}{2!}f''(0)+\frac{x^3}{3!}f'''(0)+\text{- - - - - - - -}$$

$$\log(1+x)=0+x(1)+\frac{x^2}{2!}(-1)+\frac{x^3}{3!}(2)+\text{- - - - - - -}$$

i.e. $\log(1+x)=x-\frac{x^2}{2}+\frac{x^3}{3}-\frac{x^4}{4}+\frac{x^5}{5}-\text{- - - - - - - -}$

eg: $\lim_{x \to 0} \left(\frac{\sin x}{x}\right) = ?$

$x \to 0$ ⇓ x lies in the Nbd of 0.

$= \frac{0}{0}$ form $= \lim_{x \to 0} \left(\frac{\cos x}{1}\right) = \cos 0 = 1$ Ans.

$= \lim_{x \to 0} \frac{\left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots\right)}{x}$

$= \lim_{x \to 0} \left(1 - \frac{x^2}{3!} + \frac{x^4}{5!} - \frac{x^6}{7!} + \cdots\right)$

$= 1 - 0 + 0 - 0 + 0 - \cdots = 1$ Ans

Q: for $x << 1$, $\coth x$ can be approximated as ?

(a) x

(b) $1/x$

(c) $1/x^2$

(d) e^x

x is very very small $\Rightarrow x^2, x^3, x^4 \ldots$ are also very very small i.e. negligible.

$$\coth x = \frac{\cosh x}{\sinh x} = \frac{e^x + e^{-x}}{e^x - e^{-x}} = \frac{1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \cdots}{x + \frac{x^3}{3!} + \frac{x^5}{5!} + \frac{x^7}{7!} + \cdots}$$

$$= \frac{1 + \text{Neglect}}{x + \text{Neglect}} \simeq \frac{1}{x}$$

Note: x lies in the Nbd of 'a' $\Rightarrow x \in (a-h, a+h)$

OR

$a-h < x < a+h$ where $h \to 0$

OR

$|x-a| < h$

OR

$x \to a$

OR

x is about a

Note: $|x-a| < h$

$\pm(x-a) < h$

$-(x-a) < h \;\&\; +(x-a) < h$

$x-a > -h \;\&\; x < h+a$

$x > -h+a \;\&\; x < a+h$

$x > a-h \;\&\; x < a+h$

$\Rightarrow a-h < x < a+h$

x is at 'a'

$\Rightarrow$ we are considering Exact Value

Taylor Series: If we want to Expand our function $f(x)$ in the Nbd of 'a' then expansion is called T.S.Expansion & it is defined as follows;

$$f(x) = \underbrace{f(a) + (x-a)f'(a)}_{\text{Linear Approximation}} + \frac{(x-a)^2}{2!}f''(a) + \frac{(x-a)^3}{3!}f'''(a) + \cdots\cdots$$

Note ① this series will provide Approx value of $f(x)$ in the Nbd of a

② Coeff of $(x-a)^n = \dfrac{f^n(a)}{n!}$

③ Here we are getting Increasing powers of $(x-a)$

④ To find Approx Value of $f(x)$ in LINEAR Form, Neglect 2nd & higher degree terms

⑤ " " " " " " Quadratic form, Neglect 3rd " " " "

Q: Expand $e^x + \sin x$ in the Increasing powers of $(x-\pi)$?

OR

Find the power series Expansion of $e^x + \sin x$ when $|x-\pi| < \epsilon$

OR

Find the coeff of $(x-\pi)^2$ in the power series Expansion of $(e^x + \sin x)$

OR

Find the Linear approximation of $(e^x + \sin x)$ when $x \to \pi$

OR

" " Quad " " of $(e^x + \sin x)$ about π

Sol: $f(x)=e^x+\sin x$, $f'(x)=e^x+\cos x$, $f''(x)=e^x-\sin x$, $f'''(x)=e^x-\cos x$ .....

$f(\pi)=e^\pi+0$, $f'(\pi)=e^\pi-1$, $f''(\pi)=e^\pi-0$, $f'''(\pi)=e^\pi+1$ .....

Now T.S. Exp of $f(x)$ in the Nbd of π is given as;

$$f(x)=f(\pi)+(x-\pi)f'(\pi)+\frac{(x-\pi)^2}{2!}f''(\pi)+\frac{(x-\pi)^3}{3!}f'''(\pi)+\text{-------}$$

$$e^x+\sin x=(e^\pi)+(x-\pi)(e^\pi-1)+\frac{(x-\pi)^2}{2!}(e^\pi)+\frac{(x-\pi)^3}{3!}(e^\pi+1)+\text{--------}$$

Ans of Part ① & ②

③ Coeff of $(x-\pi)^2=?=\frac{e^\pi}{2}$ ④ $e^x+\sin x=e^\pi+(x-\pi)(e^\pi-1)+$ Neglect

⑤ Quad. Approximation is, $e^x+\sin x=e^\pi+(x-\pi)(e^\pi-1)+\frac{(x-\pi)^2}{2!}e^\pi+$ Neglect.

GATE Linear Approximation of e^{-x} about 2 will be?

(a) $(x-2)e^{-x}$

(b) $(3-x)/e^2$

(c) $(3-x).e^2$

(d) $(2-x)e^{-x}$

$f(x) = e^{-x}$, $f'(x) = -e^{-x}$

$f(2) = e^{-2}$, $f'(2) = -e^{-2}$

Now T.S Exp of $f(x)$ in the Nbd of 2 is

$f(x) = f(2) + (x-2)f'(2) +$ Neglect

$e^{-x} = (e^{-2}) + (x-2)(-e^{-2})$

$= \frac{3-x}{e^2}$ Ans

Q: Expand $\log_e x$ in the Nbd of 1 & Hence Evaluate $\log_e(1.1) = ?$

Sol:

$f(x) = \log x$, $f'(x) = \frac{1}{x}$, $f''(x) = \frac{-1}{x^2}$, $f'''(x) = \frac{2}{x^3}$, ---------

$f(1) = \log 1$, $f'(1) = \frac{1}{1}$, $f''(1) = \frac{-1}{1^2}$, $f'''(1) = \frac{2}{1^3}$ ------

$= 0$ $= 1$ $= -1$ $= 2$ ------

Now T.S Exp of $f(x)$ in the Nbd of 1 is:

$$f(x) = f(1) + (x-1)f'(1) + \frac{(x-1)^2}{2!}f''(1) + \frac{(x-1)^3}{3!}f'''(1) + \text{----------}$$

$$\log x = (0) + (x-1)(1) + \frac{(x-1)^2}{2!}(-1) + \frac{(x-1)^3}{3!}(2) + \text{-------}$$

i.e $$\log x = (x-1) - \frac{(x-1)^2}{2} + \frac{(x-1)^3}{3} - \text{-----}$$

(ii) W.K.T. that, $\log x = (x-1) - \frac{(x-1)^2}{2} + \frac{(x-1)^3}{3} - \cdots\cdots\cdots$

It is the T.S. Exp of $\log x$ in the Nbd of 1. Result.

& 1.1 lies in the Nbd of 1 so we can take $x = 1.1$ in above

$$\log(1.1) = (1.1-1) - \frac{(1.1-1)^2}{2} + \frac{(1.1-1)^3}{3} - \cdots\cdots$$

$$= (0.1) - \left(\frac{0.01}{2}\right) + \left(\frac{0.001}{3}\right) - \text{Neglect.}$$

$$= 0.0953 \quad \text{Ans}$$

it is the Approx value of $\log_e(1.1)$

Shortcut Method → w.k. that,

$$\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots\cdots\cdots$$

$$\log(x) = \log[1+(x-1)] = (x-1) - \frac{(x-1)^2}{2} + \frac{(x-1)^3}{3} - \frac{(x-1)^4}{4} + \cdots\cdots$$

$$\log(1.1) = \underbrace{(1.1-1) - \frac{(1.1-1)^2}{2} + \frac{(1.1-1)^3}{3}} - \cdots\cdots$$

$$= 0.0953 + \text{Neglect.}$$

THANK - YOU

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